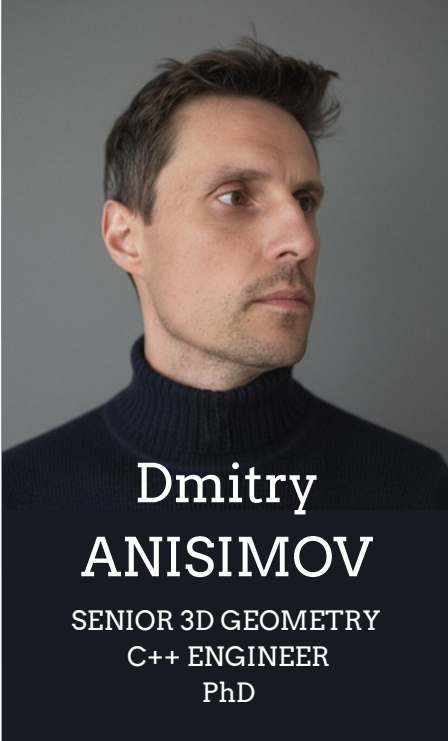




Dmitry ANISIMOV

SENIOR 3D GEOMETRY
C++ ENGINEER
PhD

Computational Geometry
Point Clouds · LiDAR
Large-Scale Processing
3D Algorithms · 3D AI

Paris, France · **Authorized to
work in France / EU**

English C1 · French B2
Italian B1

CGAL Contributor

**Google Summer of Code
Mentor** ([2018](#) · [2019](#) · [2021](#))

10+ Years Experience
2+ Years Tech Leadership
Academia · Industry · Open
Source · Startups · BigTech
Publications

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Website:
anisimov.work

PROFILE

Senior C++ / 3D Geometry Engineer with 10+ years of experience building large-scale 3D and geometry-processing systems, from research and algorithms to production. MSc in Applied Mathematics and PhD in Informatics, with expertise spanning computational geometry, large-scale point-cloud processing, LiDAR, mesh processing, shape reconstruction, shape simplification, geometry kernels, and 3D AI. Experienced across research, open source, startups, and large-scale technology products.

CORE EXPERTISE

▶ **C++ Algorithms:** modern C++, 3D algorithms & spatial data structures, computational geometry, large-scale geometry processing, performance optimization, software architecture, parallel computing, numerics

▶ **3D Geometry:** point clouds, LiDAR, mesh reconstruction, mesh processing and simplification, LODs, geometry kernels, semantic segmentation, classification, object detection, perception, Gaussian splats, 3D AI

▶ **Tech Leadership:** architecture ownership, technical roadmap, code reviews, mentorship, hiring and interviewing, team development, requirements definition, prioritization, R&D strategy, research-to-production

WORK EXPERIENCE

SAMP (acquired by **AIZE**) — 3D Geometry Lead

France · 2022–Present

- developed C++ in-house back-end large-scale point-cloud processing algorithms for areas up to tens of billion points each
- worked on semantic classification, segmentation, and object detection using geometry processing approaches
- worked on pre-processing, post-processing and various tooling required for 3D AI point-cloud processing and algorithms
- developed and maintained a backend geometry kernel supporting production 3D processing pipelines with min external dependencies (primitives, algorithms, data structures, etc.)
- turned point clouds into meshes, CAD models and supported the corresponding formats (IFC, STP, etc.)
- was in charge of the team of geometry engineers providing them with the technical support, roadmap, design guidelines and helping them to deliver various production features

ZENLY (owned by **SNAP INC**) — Senior 3D Geometry Engineer

France · 2021–2022

- developed large-scale geometry-processing algorithms for interactive 3D world maps (mesh processing with LODs, tiling)
- worked on procedural 3D model generation (buildings, bridges, road networks, terrain, and rivers), simulation (car traffic) and geometry-kernel infrastructure using in-house and external dependencies
- optimized all real-time geometry algorithms in our 3D engine to have min latency for all, including old versions, mobile devices

GEOMETRYFACTORY (owns **CGAL**) — 3D R&D Engineer

France · 2019–2021

- developed production-quality C++ computational-geometry algorithms and software for commercial geometry applications

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- contributed multiple packages to the Computational Geometry Algorithms Library (CGAL), including Shape Detection, Shape Regularization, Generalized Barycentric Coordinates (2D and 3D) and Weight Interface
- worked on mesh simplification, surface reconstruction and multi-scale geometry / LOD generation
- collaborated with researchers and engineers to turn geometric algorithms into robust, reusable open-source software

INRIA (team **TITANE**) — 3D R&D Engineer

France · 2017–2019

- developed a fully automatic large-scale 3D urban reconstruction pipeline from LiDAR point clouds
- designed algorithms for 3D reconstruction and multi-scale level-of-detail (LOD) generation
- worked at the intersection of computational geometry, point-cloud processing and large-scale spatial data
- contributed to CGAL (open source Computational Geometry Algorithms Library)

EDUCATION

USI (Università della Svizzera italiana) — PhD in Informatics

Switzerland · 2012–2017

Thesis: generalized barycentric coordinates and their applications to geometry processing

Supervisor: Kai Hormann

SPBU (Saint Petersburg State University) — MSc in Applied Mathematics

Russia · 2005–2010

Diploma: solving NP-complete problems by means of DNA properties with applications to DNA computers

Supervisor: Igor. L. Bratchikov

SELECTED PUBLICATIONS

- **Generalized Barycentric Coordinates in Computer Graphics and Computational Mechanics**, 2018 — book chapter, Editors: K. Hormann and N. Sukumar
- **Behaviour of Exponential Three-Point Coordinates at the Vertices of Convex Polygons**, Journal of Computational and Applied Mathematics, 2019, with K. Hormann and T. Schneider
- **Blended Barycentric Coordinates**, Computer Aided Geometric Design, 2017, with D. Panozzo and K. Hormann
- **Subdividing Barycentric Coordinates**, Computer Aided Geometric Design, 2016, with C. Deng and K. Hormann